

Multi-Level Information Retrieval Augmented Generation for Knowledge-based Visual Question Answering

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Abstract

The Knowledge-Aware Visual Question Answering about Entities task aims to disambiguate entities using textual and visual information, as well as knowledge. It usually relies on two independent steps, information retrieval then reading comprehension, that do not benefit each other. Retrieval Augmented Generation (RAG) offers a solution by using generated answers as feedback for retrieval training. RAG usually relies solely on pseudo-relevant passages retrieved from external knowledge bases which can lead to ineffective answer generation. In this work, we propose a multi-level information RAG approach that enhances answer generation through entity retrieval and query expansion. We formulate a joint-training RAG loss such that answer generation is conditioned on both entity and passage retrievals. We show through experiments new state-of-the-art performance on the VIQuAE KB-VQA benchmark and demonstrate that our approach can help retrieve more actual relevant knowledge to generate accurate answers.

1 Introduction

Knowledge-based Visual Question Answering (KB-VQA) has recently gained significant attention as a challenging yet promising task for evaluating systems' capabilities to deeply understand both visual and textual information in order to answer multimodal queries. In contrast to standard VQA tasks, KB-VQA further extends the challenge by requiring access to external knowledge sources (e.g., knowledge bases, knowledge graphs), as image and text queries do not explicitly carry the required knowledge for accurate answers. Recently, Large Language Models (LLMs) e.g., Palm (Chowdhery et al., 2023), LLaMA (Touvron et al., 2023) and Large Multimodal Models (LMMs) e.g., GIT (Wang et al., 2022a), BLIP-2 (Li et al., 2023), PaLi (Chen et al., 2022) and

BEiT-3 (Wang et al., 2022b) achieved impressive results in vision-language tasks thanks to their billions of parameters trained on large-scale corpora, endowing them with huge memorization capabilities for solving any downstream tasks (Izacard et al., 2023). However, they still struggle to successfully address the challenges related to knowledge-intensive tasks (e.g. KB-VQA), facing problems such as Hallucination and outdated parametric knowledge (Kandpal et al., 2023; Gao et al., 2023). Indeed, solving KB-VQA requires addressing two tasks, namely Retrieval and Reading Comprehension: relevant information (passages, documents) is first retrieved from a Knowledge Base (KB) and then answers are extracted/generated. Decoupling these two steps during training accounts for the main drawback as the retriever lacks feedback from the reader. Retrieval training is often performed using pseudo-labels indicating if the retrieved items are relevant for answering queries. For example, a common approach for passage retrieval consists in training dual encoders using pseudo-relevant passages where relevance is assessed based on string matching between passages and answers. Obviously, such heuristics yield noisy supervision retrieval signals that do not necessarily help answer questions. Retrieval Augmented Generation (RAG) (Lewis et al., 2020) has emerged as a new paradigm where generated answers serve as training signals for the retriever, circumventing the need for exhaustive memorization within model parameters and addressing the limitation of training the retriever and the reader in a decoupled way. In this work, we rely on the RAG framework to tackle the Knowledge-Aware Visual Question Answering about Entities (KVQAE) task (Shah et al., 2019; Lerner et al., 2022), which involves answering visual questions related to named entities specified in a knowledge base. Applying RAG to solve the KVQAE task involves retrieving relevant passages used as query

context during answer generation. However, the single-step retrieval employed in standard RAG may produce irrelevant passages misleading the generation process. In such a context, we found that determining which entity a question is related to plays a crucial role in generating accurate answers. We thus propose a multi-level information RAG (MiRAG) approach that performs retrieval at different levels of granularity as follows: 1) Given an initial question, we perform entity retrieval obtaining coarser-grain information which might enhance the subsequent finer-grain retrieval. 2) Using entity information, we perform query entity expansion on the initial question before applying passage retrieval offering more relevant context that improves the generator’s ability to provide accurate and informative answers. To avoid propagating irrelevant information during the iterative retrieval process, we formulate a RAG loss to train all the components in a seamless end-to-end way, such that answer generation is conditioned on both entity and passage retrieval.

2 Related work

In recent years, retrieval-augmented models have made substantial progress, driven by the necessity to exploit world knowledge for tasks like visual question answering (VQA) and open-domain question answering. Beyond the implicit knowledge captured in pretrained language models (PLMs) parameters, the retrieval augmented (RA) learning paradigm has shown how PLMs can greatly benefit from external knowledge augmentations, thus alleviating problems such as *hallucinations* (Lewis et al., 2020; Izacard et al., 2023). RA learning principle consists in optimizing through backpropagation, standard language modeling objectives combined with a retrieval step that learns providing useful knowledge during training, inference, and finetuning on downstream tasks. Regarding the Open-domain Question Answering (Open-QA) task, early work such as ORCA (Lee et al., 2019) proposed an end-to-end approach that jointly trains retriever and reader models using Inverse Cloze Task (ICT). Similarly, REALM (Guu et al., 2020) is pretrained on masked language modeling (MLM) including a differentiable retrieval step that augments an encoder with external knowledge. These examples have paved the way for more recent retrieval augmented generation (RAG) approaches that employ generative models

for reading comprehension (answer generation) instead of span-based answer extractors. For example, (Izacard and Grave, 2020) proposed an iterative process where the cross-attention scores computed for answer generation are distilled as feedback to train the retriever. (Singh et al., 2021) introduced a differentiable training method for RA-Open-QA allowing information fusion from multiple retrieved documents during answer generation. (Paranjape et al., 2021) addressed equally pseudo-relevant passages using an additional guide retriever during training that leverages the target outputs to retrieve the actual relevant passages. In contrast, (Lee et al., 2021) proposed a single model trained end-to-end which integrates retriever and reranker models as internal passage-wise attention mechanisms within a transformer architecture. While neural retrievers require large training datasets to achieve good performance, (Ram et al., 2021) explored self-supervised pre-training by recurring spans across passages in a document to create pseudo examples. In contrast, (Hofstätter et al., 2022) proposed to filter noisy training examples using confidence scores on the relevance labels, by measuring the connection between query-answer examples and items in a knowledge base. Furthermore, (Izacard et al., 2023) explored with ATLAS, a pre-trained RA language model, several objectives (e.g. likelihood distillation, attention distillation) on QA and Fact-checking downstream tasks in the few-shot setting. Other approaches dynamically perform retrieval showing that the retrieved items are not systematically beneficial for the generator. Self-RAG (Asai et al., 2023) for example generates *reflection* special tokens used during inference for retrieval decision making while FLARE (Jiang et al., 2023) employs next sentence generation token probability as a threshold to trigger the retrieval process.

Multimodal RAG. (Lin and Byrne, 2022) tackled the KB-VQA task which requires retrieval of external knowledge to answer, and proposed RA-VQA, a joint training scheme that integrates differentiable multimodal Passage Retrieval with answer generation, enabling end-to-end training. Alternatively, REVEAL (Hu et al., 2023), extended the RAG architecture with a memory module to encode multimodal world knowledge which helps retrieve relevant entries to answer queries. Additionally, they proposed an attentive fusion layer enabling seamless training of the retriever and the

generator. Finally, (Lin et al., 2024) extended RA-VQA (Lin and Byrne, 2022) with Fine-grained Late-interaction Multi-modal Retrieval (FLMR), an efficient retrieval method that increases the interaction between query and document representations using multi-dimensional embeddings. These approaches perform knowledge retrieval at the passage level only. We explore in this work how leveraging entity-level information through retrieval and query expansion may enhance RAG training for KB-VQA tasks.

3 Method

Our MiRAG proposed approach relies on the RAG framework to solve the KVQAE task which involves end-to-end training of the retriever and reader components. Figure 1 illustrates the main steps of our approach. The core idea is to perform retrieval at different levels of granularity (entity and passage) given a multimodal query with both textual and visual inputs. This multi-level retrieval strategy progressively refines the query before generating answers using a generator model. Retrieval is initially conducted at a coarse “entity” level to identify a set of candidate entities relevant to the query. Subsequently, these retrieved entities are prepended to the query before conducting the retrieval again at a finer “passage” level. Adding entities to queries provides a better understanding of the context of the query, making it easier to select more relevant passages and providing additional context for the answer generator. Formally, we aim to learn the probability $p(a \mid q, z, e)$ of generating an answer a conditioned on the query q , a retrieved passage z , and its corresponding retrieved entity e .

3.1 Entity retriever

Knowledge-intensive tasks such as KVQAE rely on an external unstructured KB (e.g. Wikipedia) from which relevant information is retrieved. Given a general-domain knowledge base represented as a set of multimodal documents $\mathcal{D}$, we define the corresponding set of entities $\mathcal{E} = \{e_i\}_{i=1}^{N_e}$, where each entity e_i is related to a document $d_i \in \mathcal{D}$, and the objective is to retrieve the most relevant entity given an input visual query. Although KVQAE questions are multimodal, we leverage only their visual content as mapping question texts to entities remains inefficient. We therefore perform cross-modal entity retrieval using a pre-

trained CLIP-based dual encoder (Radford et al., 2021) which excels at aligning image-text pairs. More formally, we encode the image of a question q and the title description of an entity into dense vector representations¹ using respectively the visual encoder $\mathbf{E}_e^i(\cdot)$ and the textual encoder $\mathbf{E}_e^t(\cdot)$. We then build a Faiss index (Johnson et al., 2019) that maps each entity $e_i \in \mathcal{E}$ to a dense vector, allowing us to perform fast exact maximum inner product search (MIPS). Formally, we compute the inner product between the dense vectors of q and all $e_i \in \mathcal{E}$ as follows :

$$s(q, e_i) = \mathbf{E}_e^i(q)^T \mathbf{E}_e^t(e_i). \quad (1)$$

During training, we compute the retrieval joint probability over the retrieved entities using the *softmax* function allowing the selection of the K most relevant entities:

$$p_\Psi(e_k \mid q) = \frac{\exp(s(q, e_k))}{\sum_{j=1}^K \exp(s(q, e_j))} \quad (2)$$

where Ψ denotes the model parameters of the CLIP visual encoder $\mathbf{E}_e^i(\cdot)$. In contrast, the CLIP textual encoder parameters are frozen to avoid recomputing embeddings of all entities in $\mathcal{E}$ at each training step. As demonstrated in (Lewis et al., 2020), combining a pre-computed index and a trainable query encoder is sufficient for RAG learning.

Achieving decent cross-modal entity retrieval performance requires strengthening alignments between visual and textual entity descriptions. Thus, we pretrain the aforementioned CLIP encoders on the cross-modal retrieval task (see Appendix B for details).

3.2 Query entity expansion

Before retrieving relevant passages useful for answer generation, we propose to expand the textual query with the top- K candidate entities retrieved in the previous stage. Query expansion techniques enrich original queries with additional context that helps improve retrieval performance, especially in settings where only pseudo-relevant supervision signals are available. Specifically, during training, given a question text q , a set of candidate entities $\{e_k\}_{k=1}^K$, we expand q by appending its text with the corresponding title of each entity e_k yielding K expanded queries ready for the subsequent retrieval.

¹We take the 512-d $L2$ -normalized hidden state vector of the start special token.

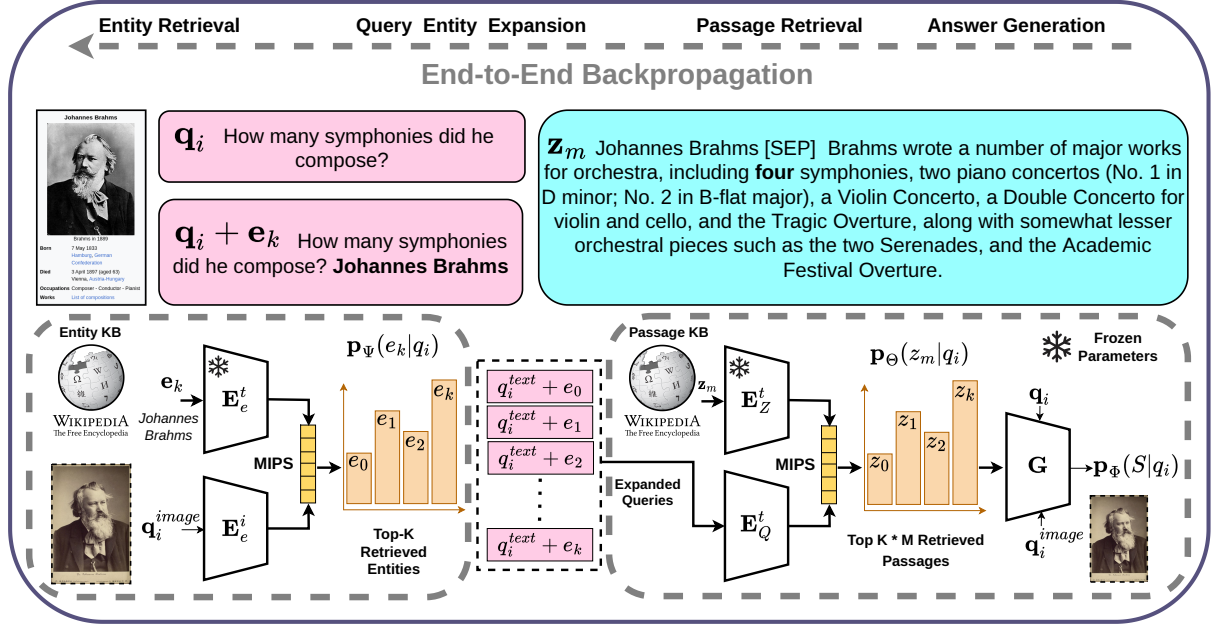


Figure 1: MiRAG approach overview.

3.3 Passage retriever

Starting from the set of documents $\mathcal{D}$, we split each document into passages of 100 words, and each passage is headed with its corresponding document title. We perform passage-level retrieval using a BERT-base (Devlin et al., 2019) dual encoder similar to the Dense Passage Retriever (DPR) (Karpukhin et al., 2020). Questions and passages are separately encoded into dense vector representations² using respectively a question encoder $\mathbf{E}_Q^t(\cdot)$ with trainable parameters Θ and a passage encoder $\mathbf{E}_Z^t(\cdot)$ with fixed parameters. Given a set of expanded questions $\{q_k\}_{k=1}^K$ and a collection of passages $\{z_i\}_{i=1}^{N_z}$, the objective is to retrieve passages that are relevant for question answering. We perform maximum inner product search for each expanded question q_k after encoding and indexing all the passages as follows:

$$s(q_k, z_i) = \mathbf{E}_Q^t(q_k)^T \mathbf{E}_Z^t(z_i). \quad (3)$$

Similar to the entity retrieval stage, we select the M highest relevance scores for each q_k , and compute the joint probability distribution of the $K \times M$ retrieved passages:

$$p_\Theta(z_i | q_k) = \frac{\exp(s(q_k, z_i))}{\sum_{j=1}^{K \times M} \exp(s(q_k, z_j))} \quad (4)$$

²We used the 768-dimensional vector of the [CLS] token from the last hidden layer of each encoder.

3.4 Answer generator

Answer generation is performed using an encoder-decoder based model denoted $\mathbf{G}(\cdot)$ with parameters Φ . We use concatenation to augment questions with the retrieved passages before feeding $\mathbf{G}(\cdot)$. The answer generator is trained on the following cross-entropy loss:

$$\mathcal{L}_{\text{Gen}} = \sum_{i=1}^{N_q} \sum_{k=1}^{K \times M} (\log(p_\Phi(s_k^* | q_i, z_k))) \quad (5)$$

Where N_q is the number of questions in a batch, s_k^* is the ground truth answer string, and $p_\Phi(s_k^* | q_i, z_k) = \mathbf{G}(q_i, z_k)$ is the probability distribution of the generated answer.

3.5 MiRAG joint training

During training, each original question q is expanded with K entities obtained after entity retrieval, then M passages are retrieved for each expanded question yielding the sets of candidate entities and passages $\{e_k\}_{k=1}^K$, and $\{z_m\}_{m=1}^{K \times M}$. Answers are generated for each z_r and the best answer a^* is selected according to the joint probability of entity retrieval, passage retrieval, and answer generation such that:

$$a^* = \arg \max_{a, z_m, e_k} p_{\Phi, \Psi, \Theta}(a | q, e_k, z_m) =$$

$$\arg \max_{a, z_m, e_k} [p_\Psi(e_k | q) \cdot p_\Theta(z_m | q) \cdot p_\Phi(a | q, z_m)]$$

(6)

In this way, we update the answer generation loss in Equation 5 such that our end-to-end learning objective guides the generator on providing answers conditioned on both the retrieved entities and passages. Thus, gradients are propagated through $G(\cdot)$, $E_e^i(\cdot)$ and $E_Q^t(\cdot)$, allowing to jointly adapt entity and passage retrievals to select the most relevant knowledge for question answering.

4 Experimental Setup

4.1 Datasets

We conduct experiments on the ViQuAE (Lerner et al., 2022) KVQAE benchmark, whose questions are specifically related to named entities (2,397 unique entities) and cover a broad range of entity types beyond named persons only, such as in (Shah et al., 2019). ViQuAE provides manually annotated challenging questions, 95.2% of which (VS 29.2% for OK-VQA benchmark (Marino et al., 2019)) require external knowledge to answer (Chen et al., 2023). ViQuAE also relies on an external Knowledge base built from Wikipedia dumps comprising 1.5M entities which act as distractors during retrieval (details in Appendix C).

4.2 Evaluation

Systems are evaluated on their ability to answer multimodal questions. For fair comparison and reproducibility purposes, we evaluate our approach using the standard metrics used for assessing the ViQuAE benchmark: F1-score and Exact Match metrics. We further evaluate our approach on passage and entity retrieval using Precision@1 (P@1), Precision@20 (P@20), and Mean Reciprocal Rank at 100 (MRR) metrics. Following (Lerner et al., 2024; Lin et al., 2024), we use pseudo-relevance to assess passages retrieval i.e., passages are considered relevant if they contain ground truth answers.

4.3 Baselines

To demonstrate the effectiveness of our proposed approach on KB-VQA, we compare its performance against published baselines in the literature and two RAG baselines settings we carefully implemented and categorized as follows. To assess the model size effect, we experimented with two pre-trained generators: T5-large (Raffel et al., 2020) with 738M parameters and BLIP2-Flan-T5-XL with 3.9B parameters. Experiment training details are given in Appendix A.

Literature baselines. Several works addressed the KB-VQA task following the two-step retrieve and read principle. (Lerner et al., 2023) proposed ECA (Early Cross-Attention) and ILF (Intermediate Linear Fusion) early fusion retrieval approaches using multimodal dense representations. (Lerner et al., 2024) proposed DPR_{v+t} , a Score-Based Fusion (SBF) approach that achieved strong KB-VQA performance where scores from text retrieval, image retrieval, and cross-modal retrieval are fused using linear interpolation combined with span-based answer extraction. Another baseline includes the PaLM LLM (Chowdhery et al., 2023) (540B parameters) using only textual queries. We do not include in our study related works that are not directly comparable. Specifically, Lin and Byrne (2022) rely on image-to-text transformations to leverage query visual content during retrieval which necessitates advanced preprocessing such as objects/attributes detection and OCR, while Lin et al. (2024) focus on improving retrieval using multi-embedding retrieval. Our approach is thus complementary to these latter.

RAG. Similar to (Lewis et al., 2020), we jointly train the answer generator and a DPR retriever following Equation 5. These baselines leverage only textual information during retrieval.

SBF-RAG. Since our approach requires visual and textual query information during entity and passage retrievals, we implemented a strong baseline referred to as SBF-RAG, which leverages, during retrieval, multi-modal information for both questions and passages. During training, the answer generator is fed with passages obtained after fusing the scores from DPR, CLIP mono-modal image, and CLIP cross-modal retrievals. Before fusion, scores are normalized to zero mean and unit variance to have comparable distributions. We used the pretrained CLIP model (Radford et al., 2021) to encode the texts and images associated with questions and passages. For text retrieval, we used the same DPR model across all experiments and settings.

5 Results

Table 1 shows the performance results of our approach and baseline systems under different settings evaluated on the ViQuAE test set. Experimental settings include whether joint training is enabled i.e., answer generator and retriever are trained end-to-end. It also includes the number

Method	Joint Fine-tuning	M_{train}	K_{train}	Metrics	
				EM	F1
Literature Baselines					
PaLM (few-shot) (Chen et al., 2023)	✗	✗	✗	31.5	-
ECA (Lerner et al., 2023)	✗	✗	✗	20.6	24.4
ILF (Lerner et al., 2023)	✗	✗	✗	21.3	25.4
DPR _{V+T} (Lerner et al., 2024)	✗	✗	✗	30.9	34.3
Systems w/o KB retrieval augmentation					
T5-large	✗	✗	✗	18.7	22.5
BLIP-2	✗	✗	✗	14.5	21.6
RAG Systems					
RAG (T5-large)	✓	5	✗	22.1	26.1
RAG (T5-large)	✓	15	✗	22.5	26.6
RAG (BLIP-2)	✓	5	✗	30.6	34.4
Score-based Multimodal fusion RAG Systems					
SBF+RAG (T5-large)	✓	5	✗	25.4	29.9
SBF+RAG (T5-large)	✓	15	✗	25.6	30.3
SBF+RAG (BLIP-2)	✓	5	✗	31.1	37.4
Ours					
MiRAG (T5-large)	✓	5	3	29.8	34.1
MiRAG (BLIP-2)	✓	2	2	36.6	41.2

Table 1: KB-VQA performance results of our approach and baseline systems evaluated on ViQuAE test sets. At inference time, the number of retrieved passages $M_{test} = 5$ for T5-large models and $M_{test} = 3$ for BLIP2 models to fit in one GPU. For systems with entity retrieval, we select the best-retrieved entity ($K_{test} = 1$) for query expansion.

of training retrieved passages M_{train} , and which type LLM/LMM is used for answer generation. In the MiRAG setting, it involves the number of training retrieved entities K_{train} so that the total number of retrieved passages is $K_{train} \times M_{train}$. Note that we systematically evaluate the different approaches in settings where the total number of retrieved passages is comparable to ensure a fair comparison. Overall, in comparable settings, we can observe that our approach consistently outperforms all baselines regardless of model size which validates the ability of MiRAG to better augment answer generators with actual relevant information that boosts the performance of KB-VQA. In particular, MiRAG (BLIP-2) with 3.94B parameters achieves the best overall performance reaching an EM of 36.6 and an F1 of 41.2, surpassing all literature baselines including PaLM (540B). In contrast to T5 models, BLIP-2 is a generator that encodes text-image input pairs for answering multimodal questions thus allowing MiRAG (BLIP-2) to benefit from text-image information at both the retrieval and answer generation steps. This suggests that even such a large multimodal model benefits from entity-level knowledge not provided by visual/textual features.

5.1 RAG joint training

We observe that RAG methods consistently enhance the performance across different models and settings compared to fine-tuned generators. RAG (T5-large) with $M_{train} = 5$ improves EM from 18.7 to 22.1 (+3.8% gain) and F1 from 22.5 to 26.1 (+3.6% gain). Increasing the number of retrieved passages M_{train} to 15 naturally provides a slight gain of 0.5% in EM (22.5) and F1 (26.6) since the answer generator benefits from more retrieved passages with potentially useful information during training. This confirms the importance of retrieval augmentation for answer generation.

5.2 Text retrieval vs multimodal retrieval

The SBF+RAG (T5-large) baseline with $M_{train}=5$ significantly improves over RAG (T5-large) achieving an EM of 25.6 (+3.5%) and an F1 of 29.9 (+4.2%) while increasing M_{train} to 15 shows only slight gains over $M_{train}=5$. Compared to the DPR-based RAG approach, the SBF+RAG approach processes multimodal information allowing better retrieval performance which in turn benefits answer generation. MiRAG (T5-large) with $M_{train}=5$ and $K_{train}=3$ further improves the performance yielding an EM of 29.8 (+4.2%) and an F1 of 34.1 (+3.4%). Similarly, our approach achieves performance gains of the same

order using BLIP-2 (3.9B params). In contrast to SBF+RAG approaches which require leveraging the image-text pairs associated with both questions and passages and combining text retrieval, image retrieval, and cross-modal retrieval, our MiRAG approach leverages only the visual query to perform cross-modal entity retrieval and textual information for passage retrieval. This suggests that expanding queries with entities helps retrieve more actual relevant passages thus validating our statement that entity-level information carries additional signals that complements visual and textual features.

5.3 Effect of increasing M_{train} and K_{train}

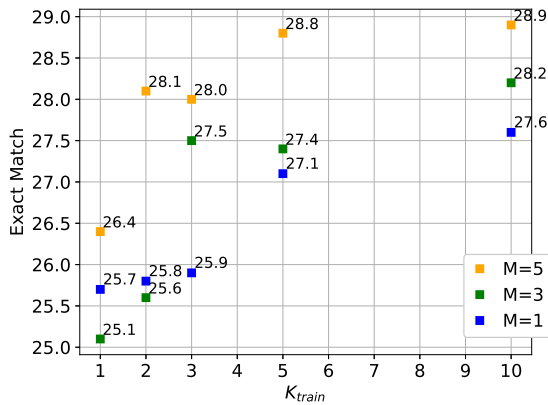


Figure 2: Exact match on ViQuAE validation set VS number of retrieved entities K_{train} and passages M_{train} using MiRAG (T5-large). At inference time, we select the best entity ($K_{test} = 1$) and feed the generator with the concatenated top-5 passages ($M_{test} = 5$). Passage concatenation showed in experiments marginally better results than selecting the passage with the best answer probability score.

Figure 2 shows the impact of increasing the number of retrieved entities and passages during Training. In general, more retrieved entities and passages increase the EM score up to some extent but at an additional memory and computational cost. For example, we found in our experiments that $K_{train} \times M_{train} = 15$ is a good trade-off between the computational cost and answer generation performance, particularly seeing that increasing the number of retrieved passages yields only marginal improvement or hurts the performance. Another key observation includes that retrieving a few entities (e.g. $K_{train}=3$) is sufficient to boost KB-VQA performance. This is confirmed in Table 2, where given a question, its correct answer

can be found in pseudo-relevant passages that are related to 4 entities on average. This highlights the importance of entity retrieval pretraining to leverage the most useful entity-level knowledge during RAG training.

Mean	Median	StdDev	Min	Max
4.00	4.00	5.45	2	60

Table 2: ViQuAE dataset statistics about the number of entities covered by the pseudo-relevant passages of a question.

Method	MRR@100	P@1	P@20
BM25	19.0	13.1	6.0
DPR	31.9	22.6	15.8
CLIP _{i-i}	15.7	12.7	6.4
CLIP _{i-t}	18.0	11.9	9.6
SBF-RAG	36.0	25.9	17.9
ILF	37.3	26.8	19.1
ECA	37.8	26.7	19.5
RAG (T5-large)	32.2	23.0	16.3
EExp-DPR	38.9	30.0	22.3
MiRAG (T5-large)	38.3	30.0	22.5
MiRAG (BLIP-2)	37.6	28.9	21.3

Table 3: Pseudo-relevance passage retrieval performance results. The best performing model settings are: RAG (T5-large) with $M_{train}=15$, MiRAG (T5-large) with $M_{train}=3$ and $K_{train}=3$, MiRAG (BLIP-2) with $M_{train} = 2$ and $K_{train}=2$.

5.4 Pseudo-relevance retrieval performance

Table 3 reports that the pseudo-relevance passage retrieval performance was evaluated using Precision@20 (P@20) and Mean Reciprocal Rank at 100 (MRR@100) metrics. While only pseudo-relevant supervision signals are available for pre-training DPR and CLIP encoders, these allow assessing the effect of answer generation feedback on pseudo-relevance retrieval performance during RAG Joint-training. We compare retrieval systems involved in our KB-VQA experiments, and baseline systems found in the literature. Those latter include two Early fusion baselines ECA (Early Cross-Attention) and ILF (Intermediate, Linear Fusion) (Lerner et al., 2023) which rely on multimodal dense representations to perform retrieval. We also report the retrieval performance of CLIP-based image and cross-modal retrieval (CLIP_{i-i} and CLIP_{i-t}) used to compute the score-based multimodal fusion system (SBF). We evaluate the effect of query entity expansion on pas-

Method	Freeze $E_e^i(\cdot)$	Entity retrieval			Passage retrieval			Answer generation	
		MRR	P@1	P@20	MRR	P@1	P@20	EM	F1
MiRAG (T5-large)	✓	40.3	30.3	13.8	38.7	30.7	21.8	28.7	33.1
MiRAG (T5-large)	✗	43.9	32.9	15	38.3	30.0	22.5	29.8	34.1
MiRAG (BLIP-2)	✓	40.3	30.3	13.8	39.6	30.4	22.4	35.1	40.0
MiRAG (BLIP-2)	✗	41.9	30.5	14.7	37.6	28.9	21.3	36.6	41.2

Table 4: Ablation study on the role of entity retrieval learning in MiRAG by stopping parameters update of the entity encoder $E_e^i(\cdot)$.

sage retrieval performance by combining the entity retriever and DPR (EExp-DPR) and measure the effect of RAG training using our MiRAG approach. DPR achieves good retrieval performance (31.9 MRR@100, 22.6 P@1) slightly enhanced by RAG(T5-large) joint training (+0.3% MRR@100, +0.4%P@1). In general, leveraging visual features drastically improves the retrieval performance for all multimodal systems indicating that the image feature complements the text feature with additional useful information. Moreover, we see that query entity expansion (EExp-DPR) shows the highest pseudo-relevant retrieval performance in MRR@100 and P@1 which indicates the potential benefits of incorporating entity information in the retrieval process. We observe that MiRAG approaches do not improve pseudo-relevance retrieval performance compared to EExp-DPR. Indeed, MiRAG’s end-to-end training objective is to improve answer generation by optimizing entity and passage encoders to retrieve the most relevant knowledge items to answer a question. Thus, pseudo-relevance retrieval metrics indicate only potential improvements for answer generation, as many pseudo-relevant passages are not truly relevant and can miss-lead answer generation training. Similarly, while MiRAG (BLIP-2) achieves superior performance on the KB-VQA task compared to MiRAG (T5-large), we note that MiRAG (T5-large) and MiRAG (BLIP-2) achieve near-similar pseudo-relevant passage retrieval performance although T5 answer generator has 5 times fewer parameters. This clearly demonstrates the contribution of our approach in providing answer generation with more pseudo-relevant passages and at the same time it learns retrieving more actual relevant ones.

5.5 Joint performance analysis

We evaluate in Table 4 the effect of gradient propagation through the entity encoder $E_e^i(\cdot)$ during

MiRAG training and the interaction between entity retrieval, passage retrieval, and answer generation performances. Overall, we observe that the best answer generation performance is obtained when enabling gradient propagation for both T5-large (29.8 vs 28.7 EM) and BLIP-2 (36.6 vs 35.1 EM) which validates the contribution of our MiRAG approach. By freezing the entity encoder, answer generation feedback is only provided to the passage retrieval encoder yielding slightly better pseudo-relevance retrieval performance. In contrast, when gradient propagation is enabled all over the components, it benefits the entity retrieval performance which in turn improves answer generation. As with passage retrieval, MiRAG (T5-large) entity retrieval performance boost is more significant than MiRAG (BLIP-2) despite the superior performance of this latter on the target task. We assume that the raw BLIP-2 ability to generate answers allows us to better guide the retrieval components towards discarding more pseudo-relevant items in favor of passages and entities that are relevant for answering questions. Appendix D shows some qualitative examples.

6 Conclusion

We proposed in this paper Multi-level information Retrieval Augmented Generation (MiRAG), a fully integrated approach for end-to-end training of RAG systems augmented with retrieved items at different levels of granularity. MiRAG combines entity and passage retrieval using query expansion that helps provide useful knowledge for answer generation. Our experiments demonstrate that our training scheme which conditions answer generation training on both entity and passage retrievals increases its ability to retrieve more relevant knowledge to solve the KB-VQA task.

7 Limitations

Our approach focuses on cross-modal entity retrieval to perform query expansion. A straightforward way to improve entity retrieval is to leverage visual and textual content associated with both the questions and entities. In the same way, performing multimodal dense passage retrieval may contribute positively to our approach as shown in (Lin et al., 2024). We conducted experiments on the ViQuAE benchmark, which is relevant for KB-VQA tasks that cover a large number of entity types. This limitation could be addressed by experimenting with other benchmarks. Finally, this work is limited to integrating entity-level information in the RAG framework. Considering entity type knowledge for MiRAG might be a good start for future investigations.

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A Training Details

We initialize our entity retriever with a pre-train ViT-B/32-base CLIP dual encoder (Radford et al., 2021) and continue pre-training following section 3.1 on cross-modal entity retrieval using 90% of the entities in the ViQuAE KB and leaving the rest for validation. We pre-train the CLIP encoder for 10 epochs using a learning rate of $2e-6$. Using gradient checkpointing, we adopt a large batch size of 1,000 to improve contrastive learning (Equation 7). The best model for entity retrieval is selected according to the in-batch mean reciprocal rank on the validation set. For passage retrieval, we pre-train the BERT-base DPR (Karpukhin et al., 2020) dual encoder on the TriviaQA (Joshi et al., 2017) dataset to ensure state-of-the-art retrieval performance and then we fine-tune it on each KB-VQA dataset using pseudo-relevant contrastive learning allowing a warm start initialization for RAG end-to-end joint training. For answer generation, we experiment with two pre-trained generators: T5-large (Raffel et al., 2020) and BLIP2-Flan-T5-XL which we finetuned on ViQuAE before retrieval augmentation. Following (Lin et al., 2024), we adopt LoRA (Hu et al., 2021), a low-rank matrix decomposition approach that greatly reduces the number of LLMs trainable parameters to train BLIP2-Flan-T5-XL on a single GPU. Our implementation relies on the huggingface-PEFT package with the following configuration: $r = 8$,

$lora_alpha = 32$, $lora_dropout = 0.1$. The batch size is set to 2 due to GPU memory limit. We train end-to-end all the components using a linear-decay schedule that reduces the learning rate from $2e-5$ to 0 after 20 epochs for the answer generator, and a fixed learning rate of $1e-6$ for the question and image entity encoders. At inference time, we decode using beam-search with 5 beams. Results on test sets were reported after averaging the scores of 2 different runs initialized with random seeds. Model checkpoints were selected based on validation performance. All experiments needed only one Nvidia A100 (80G) GPU. Our implementation is based on PyTorch (Paszke et al., 2017). Pretrained Models are obtained using Huggingface and Transformers (Wolf et al., 2020). Retrieval performance evaluation is done using the ranx library (Bassani and Romelli, 2022). Faiss (Johnson et al., 2019) allowed MIPS search and vector indexing. Our code will be released upon publication.

B Entity Retriever Pretraining

We pretrain the CLIP dual encoders on the cross-modal retrieval task using the title-image pairs in the KB which enables projecting the title and image representations of an entity closer in the embedding space. Specifically, we follow a standard in-batch negative sampling strategy and optimize the following contrastive loss:

$$\mathcal{L}_{CL} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s(q, e_i)/\tau)}{\sum_{j=1}^N \exp(s(q, e_j)/\tau)} \quad (7)$$

where τ is the temperature hyperparameter controlling the level of penalties on hard negative pairs. $\mathcal{L}_{CL}$ maximizes the embedding similarity between matching pairs and minimizes the similarity otherwise.

C Dataset Statistics

Table 5 reports the ViQuAE dataset and KB statistics.

Table 5: ViQuAE dataset and KB statistics.

KB-VQA Datasets	External KB		Splits		
	#Entities	#Passages	Train	Val	Test
ViQuAE	1.49M	11.8M	1,190	1,257	1,250

D Qualitative Examples

Figures 3 and 4 show some qualitative examples.

E Supplementary Figures

Figure 5 shows the impact of increasing the number of retrieved entities and passages during Training on passage retrieval. We note that varying the retrieved entities and passages marginally affects pseudo-relevance retrieval metrics. MiRAG end-to-end training continuously selects and discards retrieved passages according to answer generation following Equation 6.

	SBF-RAG (BLIP-2)	Generated Answer: Monaco Grand Prix
	Retrieved Passage Sauber Motorsport [SEP] The season took a turn for the worse after a 4th place by Wendlinger following the tragic deaths of Ayrton Senna and Roland Ratzenberger at the San Marino Grand Prix. Just two weeks later, Wendlinger was seriously injured after crashing in practice for the Monaco Grand Prix; losing control of his car under braking for the Nouvelle Chicane. He suffered serious head injuries, which left him in a coma for weeks, and he was sidelined for the rest of the season.	
	MiRAG (BLIP-2) + frozen entity retriever	Generated Answer:The Belgian Grand Prix Predicted Entity: Ayrton Senna
	Retrieved Passage Sauber Motorsport [SEP] The season took a turn for the worse after a 4th place by Wendlinger following the tragic deaths of Ayrton Senna and Roland Ratzenberger at the San Marino Grand Prix. Just two weeks later, Wendlinger was seriously injured after crashing in practice for the Monaco Grand Prix; losing control of his car under braking for the Nouvelle Chicane. He suffered serious head injuries, which left him in a coma for weeks, and he was sidelined for the rest of the season.	
Question He was killed when competing in which Grand Prix?	MiRAG (BLIP-2)	Generated Answer: San Marino Predicted Entity: Ayrton Senna
Answer / Reference entity SAN MARINO / Formula One		
Prediction rationales All systems successfully retrieved the passage containing the answer, however only MiRAG (BLIP-2) generated the correct answer		



	SBF-RAG (BLIP-2)	Generated Answer: Franklin Pierce
	Retrieved Passage George Peabody [SEP] In 1851, when the US Congress refused to support the American section at the Great Exhibition at the Crystal Palace, Peabody advanced £3000 to improve the exhibit and uphold the reputation of the United States. In 1854, he offended many of his American guests at a Fourth of July dinner when he chose to toast Queen Victoria before US President Franklin Pierce; Pierce's future successor, James Buchanan, then Ambassador to London, left in a huff	
	MiRAG (BLIP-2) + frozen entity retriever	Generated Answer: Barack Obama Predicted Entity: Palace of Westminster
	Retrieved Passage Barack Obama [SEP] On May 25, 2011, Obama became the first President of the United States to address both houses of the UK Parliament in Westminster Hall, London. This was only the fifth occurrence since the start of the 20th century of a head of state being extended this invitation, following Charles de Gaulle in 1960, Nelson Mandela in 1996, Queen Elizabeth II in 2002 and Pope Benedict XVI in 2010.	
Question Who was the first US President to speak in this palace?	MiRAG (BLIP-2)	Generated Answer: Barack Obama Predicted Entity: Palace of Westminster
Answer / Reference entity Barack Obama / Palace of Westminster		
Prediction rationales Predicting the correct entity helps MiRAG (BLIP-2) retrieving the actual relevant passage		

Figure 3: Qualitative examples outputs comparison of: SBF-RAG (BLIP-2), MiRAG (BLIP-2) with frozen entity retriever and MiRAG (BLIP-2). These show the benefit of leveraging entity-level information to retrieve actual relevant passages.

	SBF-RAG (BLIP-2)		Generated Answer: September	
	Retrieved Passage Johnny O'Keefe [SEP] On 14 February 1975 (St Valentine's Day) at the Masonic Hall, Waverley, O'Keefe married for the second time to Maureen Joan Maricic, a 29-year-old fashion consultant. They opened a boutique, J. O'K Creations, at Paddington in 1978. O'Keefe's last public appearance was on Seven Network's "Sounds" program, taped on 30 September 1978, six days prior to his death. Johnny O'Keefe died from a drug overdose on 6 October 1978.			
	MiRAG (BLIP-2) + frozen entity retriever	Generated Answer: March		Predicted Entity: Paul O'Grady
	Retrieved Passage George Formby [SEP] On Valentine's Day 1961, seven weeks after Beryl's death, Formby and Howson announced their engagement. Eight days later he suffered a further heart attack which was so severe that he was given the last rites of the Catholic Church on his arrival at hospital. He was revived and, from his hospital bed, he and Howson planned their wedding, which was due to take place in May. He was still there when, on 6 March, he had a further heart attack and died at the age of 56.			
Question				
this man died in which month of 1959?				
Answer / Reference entity				
February / Buddy Holly	MiRAG (BLIP-2)		Generated Answer: February	Predicted Entity: Buddy Holly
Retrieved Passage Buddy Holly [SEP] Shortly after 12:55 am on February 3, 1959, Holly, Valens, Richardson, and Peterson were killed instantly when the aircraft crashed into a frozen cornfield five miles northwest of Mason City, Iowa, airport shortly after takeoff. The three musicians, who were ejected from the fuselage upon impact, suffered severe head and chest injuries. Holly's funeral was held on February 7, 1959, at the Tabernacle Baptist Church in Lubbock.				
Prediction rationales				
Training jointly the entity retriever allowed MiRAG (BLIP-2) to correctly predict the target entity, thus providing the actual relevant passage for answering.				

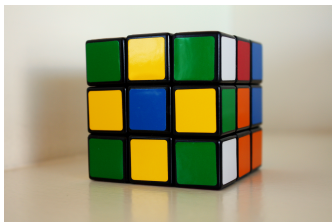
	SBF-RAG (BLIP-2)		Generated Answer: 4	
	Retrieved Passage Rubik's Revenge [SEP] However, there also exist Cubes with alternative colour arrangements (yellow opposite green, blue opposite white and red opposite orange). The Eastsheen version has purple (opposite red) instead of orange. There are 8 corners, 24 edges and 24 centres.			
	MiRAG (BLIP-2) + frozen entity retriever	Generated Answer: 6		Predicted Entity: Rubik's Cube
	Retrieved Passage Rubik's Cube [SEP] It is widely considered to be the world's best-selling toy. On the original classic Rubik's Cube, each of the six faces was covered by nine stickers, each of one of six solid colours: white, red, blue, orange, green, and yellow. The current version of the cube has been updated to coloured plastic panels instead, which prevents peeling and fading.			
Question				
How many squares does this toy have on each side?				
Answer / Reference entity				
Nine / Rubik's Cube	MiRAG (BLIP-2)		Generated Answer: 9	Predicted Entity: Rubik's Cube
Retrieved Passage Rubik's Cube [SEP] It is widely considered to be the world's best-selling toy. On the original classic Rubik's Cube, each of the six faces was covered by nine stickers, each of one of six solid colours: white, red, blue, orange, green, and yellow. The current version of the cube has been updated to coloured plastic panels instead, which prevents peeling and fading.				
Prediction rationales				
Use case where MiRAG (BLIP-2)-Frozen does not answer correctly even though it has retrieved the correct entity and the actual relevant passage				

Figure 4: Qualitative examples outputs comparison of: SBF-RAG (BLIP-2), MiRAG (BLIP-2) with frozen entity retriever and MiRAG (BLIP-2). These show the benefit of leveraging entity-level information to retrieve actual relevant passages.

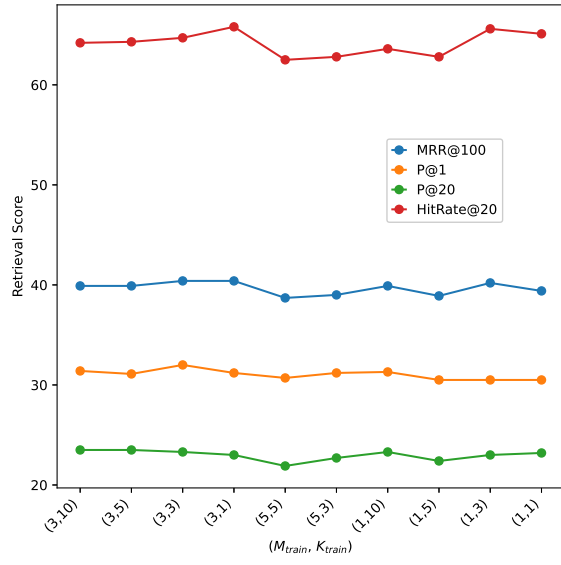


Figure 5: Pseudo-relevance passage retrieval performance VS number of retrieved entities K_{train} and passages M_{train} using MiRAG (T5-large).